

Package ‘rENM.dev’

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Type Package

Title Tools under development for the rENM framework

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Description The rENM framework is a modular sysem for reconstucting and analyzing long-term ecological niche dynamics using time-indexed environmental data and species occurrence records. Its packages support climate-based niche modeling, temporal analysis of climatic suitability, and reproducible workflows for studying ecological change.

Depends R (>= 4.1.0)

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Encoding UTF-8

Roxygen list(markdown = TRUE)

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URL <https://github.com/rENM-Framework/rENM.dev>

BugReports <https://github.com/rENM-Framework/rENM.dev/issues>

Imports rENM.core,
curl,
httr2

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assemble_ai_package *Assemble GenAI suitability package for a species*

Description

Collects selected suitability-trend rasters, summary tables, and species metadata for a single species, then builds an AI-ready package together with a prompt template and log entry. The function locates the project root via `rENM_project_dir()` and no longer relies on hard-coded paths.

Usage

```
assemble_ai_package(
  alpha_code,
  files = c("centroids/Bioclimatic-Velocity.csv",
            "centroids/Centroids-Latitude-Summary.csv",
            "centroids/Centroids-Longitude-Summary.csv",
            "suitability/Suitability-Change-Trend.tif",
            "suitability/Suitability-Trend-State-Analysis-Summary.csv",
            "suitability/Suitability-Trend.tif", "suitability/Species-Information.csv",
            "variables/Variable-Contributions-BR-Stats.csv")
)
```

Arguments

<code>alpha_code</code>	Character. Four-letter species alpha code (e.g., "CASP").
<code>files</code>	Character vector. File paths (without <code>alpha_code</code> prefix) identifying which outputs to include in the AI package. All entries must include a subdirectory relative to Trends/, for example: "suitability/Suitability-Trend.tif" or "centroids/Bioclimatic-Velocity.csv". Each file will be resolved as: <code><project_root>/runs/<alpha_code>/Trends/<subdir>/<alpha_code></code> . Defaults to a minimal core set.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by `rENM_project_dir()`.

Key update

- Adds flexible file selection via `files`.
- All files are drawn from subdirectories under Trends/.
- Users must explicitly specify subdirectories (e.g., "suitability/Suitability-Trend.tif" or "centroids/Bioclimatic-Velocity.csv").

Pipeline context

- Copies selected suitability rasters and tables into `suitability_package/`.
- Copies `\_species.csv` to a species-specific file and includes it in the package.
- Creates a `.zip` archive of the package contents.
- Copies the ChatGPT prompt template resolved via `system.file("chatgpt", "suitability_prompt.txt", package = "rENM.ai")` into the species `chatgpt/` directory.
- Appends a standardized processing-summary block to `\_log.txt`.

Value

Named List (invisible).

- alpha_code – input species code
- pkg_dir – path to suitability_package
- alpha_chatgpt_dir – path to species chatgpt folder
- zip_path – path to created archive
- prompt_path – path to copied prompt file
- species_info_trends – trends-level species table
- species_info_pkg – table inside the package
- log_file – updated log file

Examples

```
## Not run:
assemble_ai_package("CASP")

## End(Not run)
```

render_ai_docx

Render GenAI DOCX report to PDF

Description

Converts a species-specific DOCX report into PDF and appends a processing summary to the run log.

Usage

```
render_ai_docx(alpha_code, verbose = TRUE)
```

Arguments

alpha_code Character. Species alpha code, for example "CASP".

verbose Logical. If TRUE (default), prints diagnostic messages.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Inputs

- One DOCX file per species located at <rENM_project_dir()>/runs/<alpha_code>/Summaries/pages/<alpha_code>-Suitability-Trend-Analysis.docx

Outputs

- A PDF written to the same directory: <alpha_code>-Suitability-Trend-Analysis.pdf
- A multi-line processing summary appended to <rENM_project_dir()>/runs/<alpha_code>/_log.txt

Methods

- Conversion uses LibreOffice soffice in headless mode.
- File sizes and time stamps are recorded for auditing.

Value

Character. Invisibly returns the path to the generated PDF.
Side effects:

- Writes a PDF file alongside the source DOCX.
- Appends a formatted record to the species-level _log.txt.

Examples

```
## Not run:
# Convert the Cassin's Sparrow report and view the resulting PDF path
pdf_path <- render_ai_docs("CASP")

## End(Not run)
```

submit_ai_package	<i>Submit GenAI-ready climatic suitability trend package for a species</i>
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Description

Runs an end-to-end climatic suitability trend analysis for a single species by sending a zipped data bundle and custom prompt to the OpenAI Responses API, retrieving a DOCX report, and logging processing metadata and cost information.

Usage

```
submit_ai_package(
  alpha_code,
  model = "gpt-5.1",
  api_key = Sys.getenv("OPENAI_API_KEY")
)
```

Arguments

alpha_code	Character. Four-letter species alpha code.
model	Character. OpenAI model identifier.
api_key	Character. OpenAI API key.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by `rENM_project_dir()`.

Pipeline context

- Reads a species-specific prompt file.
- Uploads a zipped bundle of GeoTIFF and CSV inputs.
- Calls the Responses API with the code-interpreter tool enabled.
- Downloads a Word report created inside the container.
- Computes token usage and approximate OpenAI cost.
- Appends an eBird-style summary line to `_log.txt`.

Expected inputs

- Prompt file containing the placeholder string `<alpha_code>`.
- Zipped bundle of raster and tabular data required by the prompt.

Authentication

- Uses `api_key`, defaulting to `Sys.getenv("OPENAI_API_KEY")`.

Directory layout under `rENM_project_dir()`

```
<project_dir>/runs/CASP/Summaries/chatgpt/suitability_package.zip
<project_dir>/runs/CASP/Summaries/chatgpt/suitability_prompt.txt
<project_dir>/runs/<alpha_code>/Summaries/pages/
  <alpha_code>-Suitability-Trend-Analysis.docx
<project_dir>/runs/<alpha_code>/_log.txt
```

Outputs

- DOCX report saved to the pages directory.
- Updated `_log.txt` containing a processing summary.

Value

List (invisible) with components

- `response`
- `docx_path`
- `elapsed_sec`
- `input_tokens`, `output_tokens`, `total_tokens`
- `est_total_cost`

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